



Raksha

Every scam follows a script.
Now your phone knows the script.

The real-time scam-call shield. Script recognition in about 10 ms during a live call, powered by Moss. It interrupts before the OTP leaves your mouth, tells you what to say, and alerts someone you trust.

Shivam Gupta · with Claude · github.com/shi1720/YC-Fall-2026-x-Moss

YC Fall 2026 × Moss · Zero Latency Builder Sprint · Real-Time Voice & Conversational AI

THE PROBLEM

Scams are conversations with a script. Nobody protects people during the conversation.

₹22,845 cr

reported cyber-fraud
losses in India, 2024 —
10× the 2022 figure

MHA, Lok Sabha Q.432

₹1,900 cr

taken by 'digital arrest'
scams from 1.23 lakh
people in 2024 (+465%)

MHA · Supreme Court suo motu,
2025

41%

of \$10k+ losses by older
adults in the US began
with a phone call

FTC Data Spotlight, 2025

\$25.6M

wired by one employee
to a deepfaked CFO on a
video call

Arup, Hong Kong, 2024

Every one of these calls followed the same arc: authority → fear → secrecy → the ask.

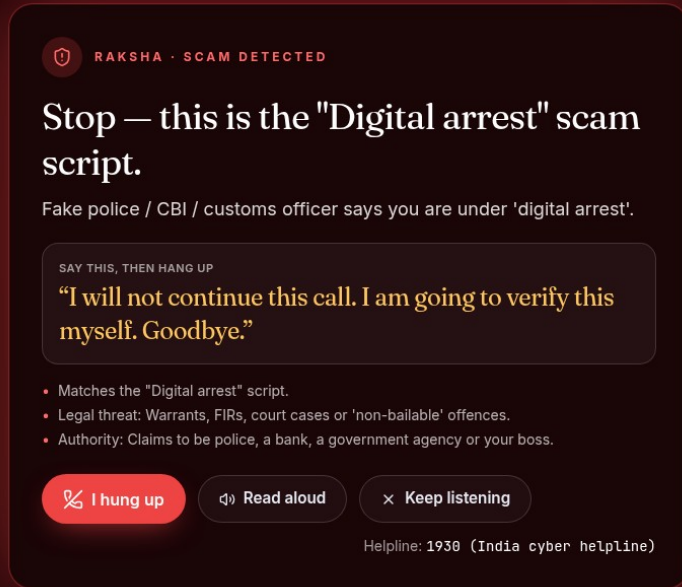
THE GAP

Caller-ID tells you who is calling. Nobody tells you what they are doing to you.

	Truecaller · Airtel · Jio · iOS 26	Number reputation	Crews rotate numbers daily. The script never changes.
	Truecaller AI Call Scanner	Synthetic-voice detection	Most scammers are humans reading a script.
	Google Pixel Scam Detection	On-device content analysis	Pixel 9+ only — under 1% of Indian phones.
	Hiya AI Phone	Cloud content analysis	US only, \$9.99/month, cloud round-trip per fragment.
	Raksha	Script-level detection + intervention	Any phone. In-process retrieval. Says what to do. Alerts family.

THE PRODUCT

Meet Raksha. It listens with you, recognises the script, and steps in.



Every fragment, even mid-sentence

Speech becomes text on the device; each fragment is a query.



Recognised in ≈ 10 ms

Matched against 409 real scam lines in-process with Moss, including on-device embedding. No vector DB, no round-trip.



One sentence to say

It names the script, gives the exact words to end the call, and speaks them aloud.

A fast path that never waits for the cloud. A slow path that knows what to say.

FAST PATH · EVERY FRAGMENT · ≈ 10 MS

- Moss runtime, in-process: the playbook index is loaded at boot and queried with raw cosine scores (alpha 1.0, top-6).
- Cosine $\rightarrow$ calibrated confidence. Benign look-alikes in the same index out-vote tactic lines (“we will never ask for your OTP” stays quiet).
- Speaker-aware: the person's own words only ever count as compliance.
- Noisy-OR over 21 persuasion tactics; evidence persists for the whole call.
- Rule 1 — pressure + ask (the triad) $\Rightarrow$ DANGER. Rule 2 — victim about to comply $\Rightarrow$ intervene now.

SLOW PATH · ASYNC · ON TRANSITIONS

- A 20B model (Groq) returns a JSON contract: verdict · explanation · say this · action.
- Only when risk changes, never per fragment: ≈ 300 –800 ms, off the critical path.
- A confident “benign” verdict can veto CAUTION. It can never override DANGER.
- Template fallback: the product works with no LLM key at all.

One Node process: Next.js, WebSockets and the Moss runtime share one memory.

✓ Raksha — system architecture

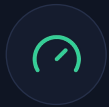
Real-time scam-call shield. One utterance in, one verdict out, in single-digit milliseconds — retrieval by Moss, in-process.



LATENCY BUDGET FOR ONE SPOKEN SENTENCE (≈ 2,500 MS)

speech → text ≈ 200–400 ms (streaming) · Moss embed + search ≈ 10–18 ms (search < 1 ms) · risk engine < 0.1 ms · WebSocket round-trip ≈ 20–80 ms · coach (async) ≈ 300–800 ms, off the critical path

Four ways the retrieval layer does real work here.



Loaded index, in-process query

raksha-playbook (409 lines) loaded at boot. Every fragment embedded and searched locally via queryMultiIndex (raw cosine, calibrated). The entire hot path ≈ 10 ms, unmetered.



Call memory in a session

Every live turn goes into one in-process Moss session tagged by call id; the guardian's "what did they ask for?" is a filtered semantic query over it. Never pushed — a call's turns are deleted when it ends.



Auto-refresh hot-swap

Reported caller lines are upserted into raksha-intel. Every running shield polls and hot-swaps the new version in with zero query downtime. No redeploy.

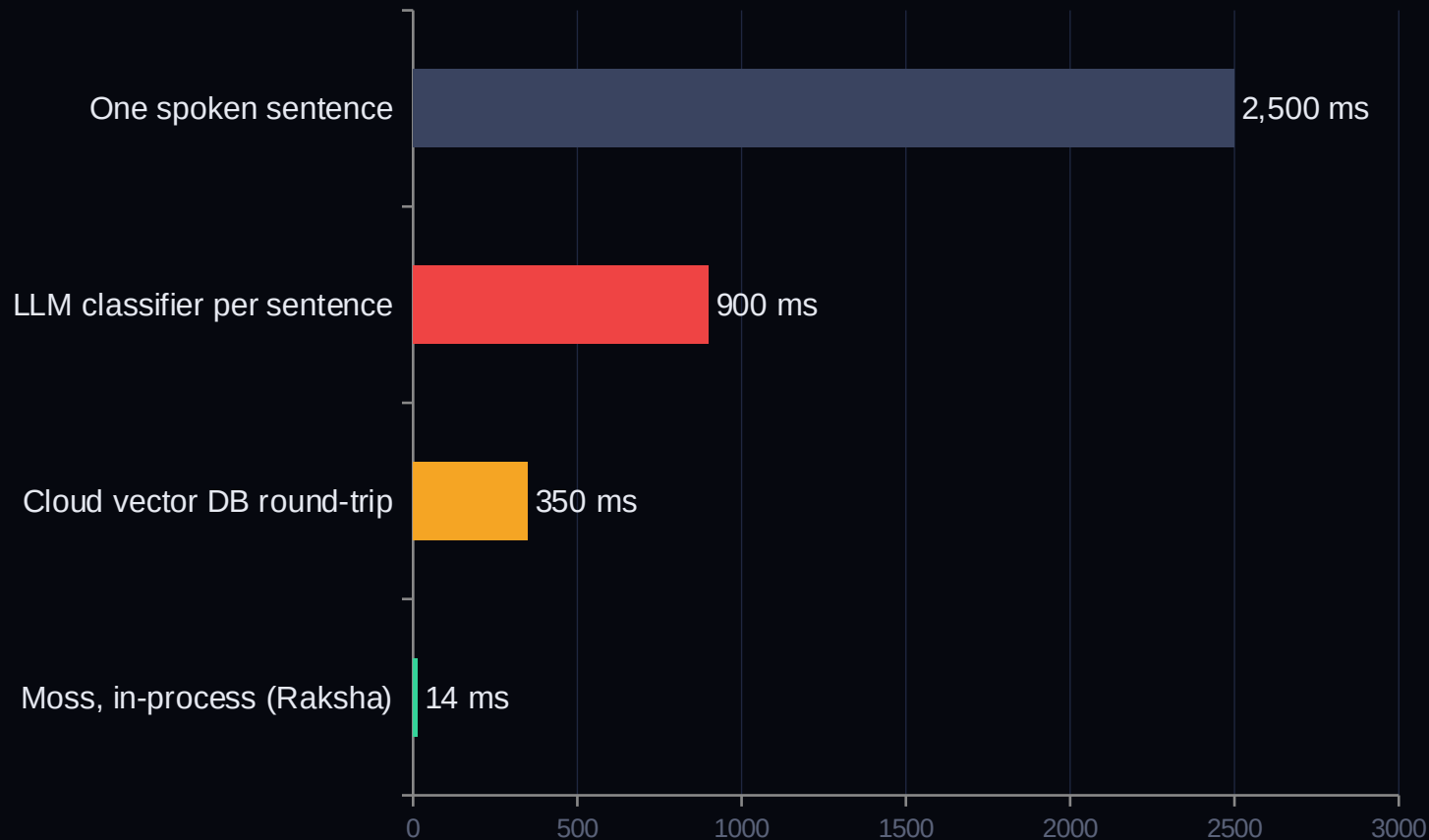


Multi-index, one top-K

queryMultiIndex searches the curated playbook and community intel together. Metadata (family, tactics, severity, kind, stage) lets the engine reason over tactics, not text.

A scam is won or lost inside one sentence.

“Read me the six digits, quickly” takes ~2.5 s to say. How much of that sentence is gone before the shield has an answer?



≈ 1%

of the sentence consumed
by Moss retrieval (p95 ≈ 27
ms measured)

14%

consumed by a 350 ms
cloud vector-DB round-trip

36%

consumed by a 900 ms LLM
classifier — the OTP is
already out

₹0

marginal cost per fragment:
local Moss queries are
unmetered

Scams isolate. Raksha breaks the isolation.



A family code, no accounts

Links a parent's phone to a daughter's dashboard. Risk, transcript highlights and interventions, live.



Speak through the shield

Her words are shown and read aloud on the phone — over the scammer's voice.



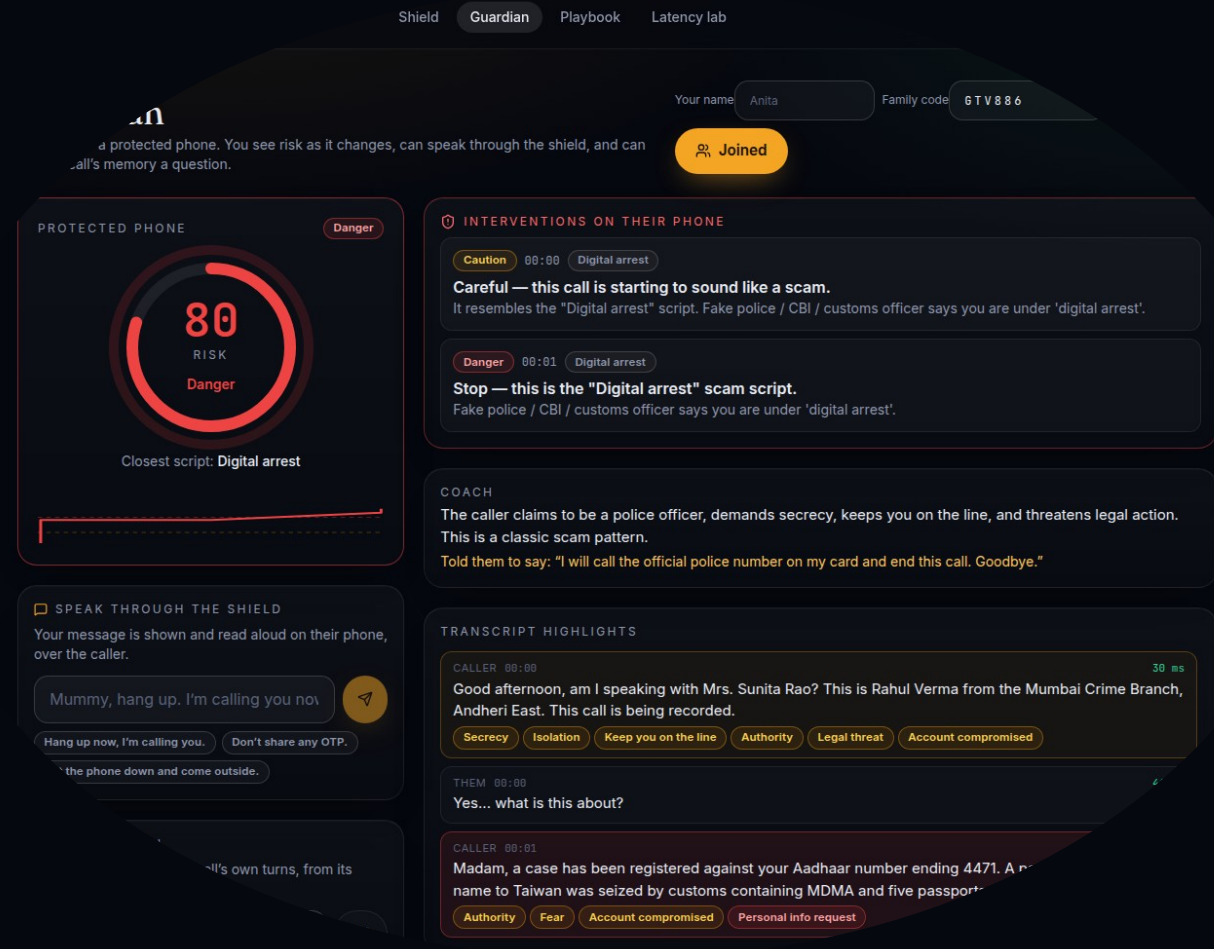
Ask the call

“What money did they ask for?” — answered from the call's Moss session in milliseconds.



Human and agent, one session

Multiplayer by design: the person, the shield and the guardian share the same live context.



Measured, not claimed.

Eighteen full call transcripts replayed through the real Moss runtime — ten scams, eight genuine calls — plus a live benchmark page anyone can run against the deployed instance.

100%

scam scenarios caught (reached DANGER)

0%

false alarms on genuine calls

90%

scripts named correctly

3.9

mean turn at which DANGER fired

8.0 ms

retrieval p50, embed + search

14.9 ms

retrieval p95

Full turn-by-turn report: [docs/eval/REPORT.md](#) · runtime: Moss, in-process

Who pays, and why the economics work.



Banks & payment apps

From 1 Jan 2027 RBI makes banks compensate scam-induced losses; UK banks already reimburse £354M/yr. A per-customer licence (₹5-15/month) that stops even a fraction of transfers beats any refund — and gives the bank a real-time “transfer hold” signal.



Telcos & OEMs

A “protected line” add-on (₹49-99/month) on any Android phone, not just a Pixel. DoT's kill-switch ambitions need exactly this signal.



Families

A ₹99/month circle that protects parents — the price band where family-safety apps already convert (Life360: 2.8M paying circles).

UNIT COST PER PROTECTED CALL

≈ ₹0.02

- Moss local queries: ₹0 (unmetered)
- STT on device: ₹0
- LLM coach, ≤ 6 calls on a 20B model: ≈ ₹0.02
- One small container serves thousands of concurrent calls

One prevented ₹1.5-lakh digital-arrest transfer pays for ~7 million protected calls.

“Truecaller tells you who is calling. Raksha tells you what they are doing to you — and what to say back.”

Every phone, not just a Pixel.

Now

Web reference implementation, deployed. 18 scenarios, live mic, recordings, guardian, community intel, eval.

+4 weeks

Android app with in-call audio; Hindi / Tamil / Telugu playbooks; bank “transfer hold” webhook.

+3 months

Hot path fully on-device with Moss mobile/browser SDKs; deepfake-voice liveness fused into the risk model; pilot with one bank and one senior-living network.

+6 months

Federated playbook across partners; one-tap 1930 / NCRP reporting; regional script variants.



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